

# AJITHSAMPATH

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## EDUCATION

**University of Geneva, Switzerland | Postdoc** **2025 to Present**

- Primarily working on Beam characterization of HIRAX array.

**University of KwaZulu - Natal, South Africa | PhD** **2021 to 2024**

- Thesis title:  
Beam Calibration and Foreground Leakage in 21cm Intensity Mapping - Application to HIRAX.  
Supervisor : Prof. Kavilan Moodley
- NRF - TWAS Doctoral Fellowship

**University of Madras, Chennai, India | MSc Theoretical Physics** **2017 to 2019**

- Related Coursework:  
Astrophysics, Theoretical Cosmology, Numerical Methods.
- Final semester short Project: HI observations of Barred Spiral Galaxies.

## PUBLICATIONS

- **Primary Beam Chromaticity in HIRAX: I. Characterization from Simulations and Power Spectrum Implications** **2024**  
Ajith Sampath, Devin Crichton, Kavilan Moodley, et al. (Submitted to The Astrophysical Journal - Preprint arXiv link : <http://arxiv.org/abs/2412.09527>)
- **The HIRAX: Overview and Status Update.** **2024**  
Anthony Walters, et al. 18th European Conference on Antennas and Propagation (EuCAP), Glasgow, UK, 2024, pp. 1-5, doi:10.23919/EuCAP60739.2024.10501639
- **Full-wave analysis of mutual coupling in the HIRAX radio telescope.** **2022**  
Q Gueuning, D Crichton, A Sampath, AK Brown, E de Lera Acedo & K Moodley. International Conference on Electromagnetics in Advanced Applications (ICEAA), Cape Town, South Africa, 2022, pp. 288-288, doi: 10.1109/ICEAA49419.2022.9899905.
- **HIRAX: 256 -element array status and overview.** **2022**  
Devin Crichton, et al. Journal of Astronomical Telescopes, Instruments, and Systems Vol. 8, Issue 1, 011019 <https://doi.org/10.1117/1.JATIS.8.1.011019>
- **Antenna characterization for the HIRAX experiment.** **2022**  
Emily R.Kuhn, et al. Ground-based and Airborne Telescopes IX. Vol. 12182. SPIE, 2022. <https://doi.org/10.1117/12.2627856>

## SKILLS

- **Technical**  
Python, CASA, Machine Learning tools, Linux and HPC experience.
- **Language**  
English (Academic IELTS 7.5 - C1)  
German and French introductory modules in MSc years.  
Tamil - Native language.

- **Interests**

21cm Cosmology, Intensity Mapping, Foreground filtering techniques, and Primary beam characterization.

## **TEACHING EXPERIENCE - AS TUTOR**

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- **Introduction to Calculus** **2023 & 2022**  
First year Undergrad course
- **Further Calculus & Introductory Real Analysis** **2023**  
Second year Undergrad course
- **Methods of Applied Mathematics** **2022**  
Third year Undergrad course

## **CERTIFICATION & AWARDS**

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- **"Biggest Impact" award for Oral Presentation** **2023**  
Postgraduate Research and Innovation Symposium, UKZN.
- **"Winner" of BRICS Astronomy Hackathon on Machine Learning** **2023**  
BRICS Astronomy meeting, Capetown.
- **Awarded PhD Fellowship** **2021**  
Funded by National Research Foundation and The World Academy of Sciences.

## **OUTREACH ACTIVITIES**

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- **Lead the Weekly KREATE Computer Literacy workshop** **2023**  
For underprivileged high school students from Lindelani Youth Forum and Community Life Centre, Durban.
- **Co-lead two KREATE Machine Learning workshops** **2023**  
For students and industry staffs hosted by SmartXchange.

## **CONFERENCES**

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### **ORAL PRESENTATIONS**

- **SKA Postgraduate Research Conference** **2023**  
*Talk Title : An Analytical approach to characterise Beam Systematics in HIRAX*  
Hosted by SARA0 in Capetown.
- **Postgraduate Research and Innovation Symposium.** **2023**  
*Talk Title: Understanding the Instrument that tries to understand the universe.*  
Hosted by University of KwaZulu-Natal (Won "Biggest impact award").
- **Cosmology on Safari** **2023**  
*Talk Title: An Analytical approach to characterise Beam Systematics in HIRAX*  
Hosted by University of KwaZulu-Natal, in Hluhluwe.
- **SKA Postgraduate Research Conference** **2022**  
*Talk Title: HIRAX Primary Beam modelling*  
Hosted by SARA0 in Vanderbijlpark.
- **Swiss - SA HIRAX bilateral workshop** **2022**  
*Talk Title: Modelling the effects of HIRAX primary beam*  
Hosted by University of KwaZulu-Natal in Magaliesburg.

- **UKZN - Swiss Mini HIRAX - MeerKAT Workshop** **2022**  
*Talk Title: Analytical Modeling of HIRAX primary beam*  
Hosted by University of KwaZulu-Natal in Durban.

## POSTER PRESENTATION

- **IAU GA Symposium** **2024**  
*Poster title: Effects of HIRAX beam chromaticity in power spectrum*  
Hosted by International Astronomical Union in Capetown.
- **BRICS Astronomy Meeting** **2023**  
*Poster title: Effects of HIRAX beam chromaticity in power spectrum*  
Hosted by SAAO in Capetown.